



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-024A FleetPulse Intake & Recovery

C3-01 competency Gate | Fleet telemetry + maintenance events

Artifact	C3_024A_FLEETPULSE_INTAKE_RECOVERY_GATE_RC1
Status	GATE A - INDEPENDENT FIRST ATTEMPT
Teaching standard	M01 beginner-first teacher loop. Videos are visual companions; the book remains sufficient fallback.
C2B boundary	Bridge foundations may be assumed, but C3 engineering behavior is demonstrated, broken, repaired, transferred and validated.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

Course rule: preserve what arrived -> validate interpretation -> quarantine ambiguity -> publish durably -> commit state last -> prove replay safety. A green job is never sufficient evidence by itself.



Gate mission

Build a small production-style ingestion/recovery solution from the supplied unseen case. Use the same concepts as C3-01 but do not copy ShopPulse field names or solution structure blindly.

NO SOLUTION HELP DURING ATTEMPT

No lesson/review/video/tutorial/AI solving during the independent Gate attempt. Normal language/tool syntax help may be used only to remember command syntax; it must not design the solution.

REQUIRED BUILD

- Declare source contract + output grain/window.
- Preserve raw case inputs and a reproducible manifest/fingerprint.
- Parse/normalize defensively with explicit quarantine or block decisions.
- Implement deterministic incremental/change/state behavior appropriate to this case.
- Inject one failure before state commit and recover by replay.
- Run independent source/output/state reconciliation and at least one edge/failure test.
- Write sanitized logs + README/runbook + 3-5 minute interview defense.

PASS RULE - COMPETENCY ONLY

Every required competency G1-G7 must PASS and critical failures must equal zero. Every required competency stands on its own; one failed competency cannot be compensated by stronger performance elsewhere.

Competency evidence table

ID	Competency	PASS evidence	Status
G1	Source contract & landing	Inputs/windows/grains declared; raw + manifest/fingerprint preserved.	PASS / FAIL
G2	Defensive parsing & evolution	CSV/JSON/schema/time behavior explicit; unsafe ambiguity quarantined/blocked.	PASS / FAIL
G3	Incremental/CDC semantics	Stable key/cursor/event semantics; ties, updates/deletes handled deliberately.	PASS / FAIL
G4	State commit & replay safety	Durable publish/validate before state; injected failure + restart proven.	PASS / FAIL
G5	Interface reliability	Auth secret boundary if relevant; timeout/pagination/retry/rate-limit rules justified.	PASS / FAIL
G6	Testing/reconciliation	Independent controls + edge/failure test; source/output/state reconciled.	PASS / FAIL
G7	Operational handoff	Sanitized logs, runbook, recovery/backfill policy, interview defense.	PASS / FAIL

CRITICAL FAILURES

- Raw evidence overwritten or unavailable without acknowledged blocker.
- State/checkpoint advances past data that was not durably validated/published.
- Unresolved schema/time/key ambiguity silently accepted.
- Replay/restart skips eligible data or causes unintended accepted duplicates.
- Real credential exposed or fabricated evidence submitted.
- Independent reconciliation knowingly fails or is omitted for a critical boundary.
- Protected Gate review opened before first attempt was frozen.



Failure/recovery and evidence freeze

MANDATORY FAILURE DRILL

Choose one realistic pre-commit failure point after at least some work has happened. Prove the committed checkpoint remains safe; inspect partial side effects; rerun; prove no eligible data was skipped and accepted/current result remains deterministic.

FREEZE ATTEMPT

Save under `evidence/GATE\_A/attempt\_001/`. Freeze source/code/config/state/log/test/reconciliation/runbook evidence before opening C3-024A review. Do not overwrite attempt_001 later.

INTERVIEW DEFENSE

Defend architecture, grain/window, raw-vs-staging boundary, state/commit order, one ambiguity, one failure, one independent control, one scaling tradeoff.

NEXT ROUTE

Gate A PASS -> C3-02. Gate A weak competency -> targeted remediation + changed micro-retest. Gate B/C are full fresh retests only when explicitly assigned; repair does not unlock them automatically.